Data Provided: Erlang C Curves



The University of Sheffield

DEPARTMENT OF COMPUTER SCIENCE

AUTUMN SEMESTER 2005–2006 **2 HOURS**

NETWORK PERFORMANCE ANALYSIS

Answer THREE questions.

All questions carry equal weight. Figures in square brackets indicate the percentage of available marks allocated to each part of a question.

1. In an $M/G/1$ queue inter-arrival times are exponentially distributed and service times are 'generally' distributed. The average wait in the queue may be derived in terms of the residual service time as follows

$$\bar{W} = \frac{\bar{R}}{1-\rho},$$

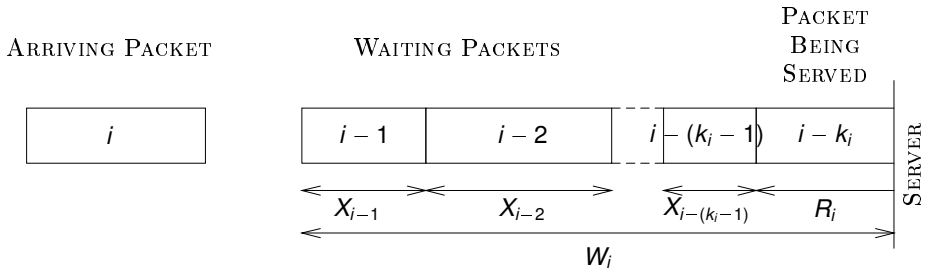
where $\bar{R}$ is the average residual service time and ρ is the queue utilization factor.

a) Describe, using diagrams where necessary, the concept of residual service time. State the assumptions used when deriving the formula above.

[30%]

Answer

The residual service time is the remaining service time as perceived by an arriving customer. The ith customer arrives and waits R_i time units for the current customer under service to finish being served. This is the residual service time. The diagram below explains this concept:



In the diagram the ith arriving packet perceives a queueing time of W_i and a residual service time of R_i. The average residual service time is the average R_i for all packets.

The assumptions used in deriving the formula are

- We assume that time averages are equivalent to steady state averages.
- We assume that service times have a finite variance.
- We assume that the arrival times are statistically independent of the service times.
- Assume the queue is FIFO.

5 marks for each assumption, 10 marks for the description of residual service time+diagram.

b) Use a graphical argument to derive a formula for the residual service time. Combine this equation with the formula in part a) for the average waiting time to show that

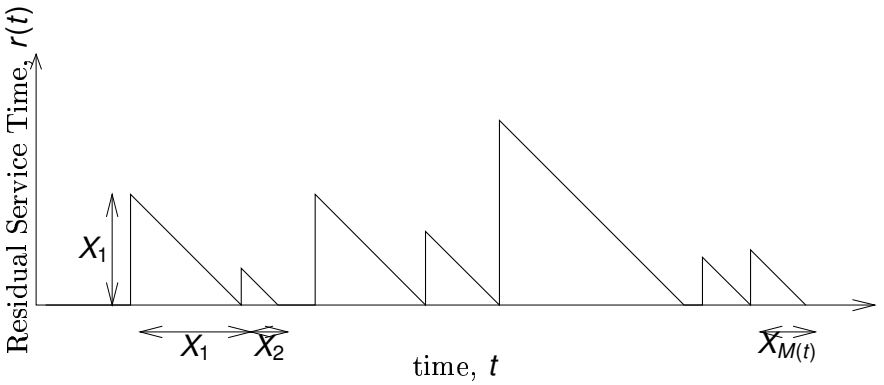
$$\bar{W} = \frac{\lambda \langle X^2 \rangle}{2(1-\rho)},$$

where λ is the arrival rate and $\langle X^2 \rangle$ is the second moment of the service time distribution.

[40%]

Answer

We again turn to a graphical proof to obtain the value of the mean residual service time. A plot can be drawn of the residual service time over time.



$M(t)$ is the number of service completions in the interval $[0, t]$.

The time average of the mean residual service time can then be calculated in terms of the area under the curve, $A(t)$, which is given by the area under each of the triangles,

$$A(t) = \frac{1}{2}\sum_{i=1}^{M(t)} X_i^2.$$

The average service time is the area divided by time, t,

$$\begin{aligned}\frac{A(t)}{t} &= \frac{M(t)}{t}\frac{A(t)}{M(t)} \\ &= \frac{1}{2}\frac{M(t)}{t}\frac{\sum_{i=1}^{M(t)} X_i^2}{M(t)}.\end{aligned}$$

where we have used the number of arrivals after time t, as $M(t)$. As $t \to \infty$, *if the variance of the service time distribution is finite*, we find that

$$\bar{R} = \frac{1}{2}\lambda\left\langle X^2\right\rangle.$$

Again, assuming that time averages and ensemble averages are equal, we substitute back into the equation provided in the previous part to obtain the Pollaczek-Khinchin formula

$$\bar{W} = \frac{\lambda\left\langle X^2\right\rangle}{2(1-\rho)}.$$

15 for the diagram, 10 for the area formula, 10 for the introduction of t and $M(t)$ and 5 for the right answer.

c) The equation you have derived is recognized as the Pollaczek-Khinchin formula. Use it to derive the average waiting time in an $M/M/1$ queue.

[15%]

Answer

In an $M/M/1$ queue there is an exponential service time,

$$p(X) = \mu \exp(-\mu x).$$

The mean of this service time is $\langle X \rangle = \frac{1}{\mu}$ and the variance is $\mathrm{var}(X) = \frac{1}{\mu^2}$. Therefore the second moment is therefore

$$\langle X^2 \rangle = \langle X \rangle^2 + \mathrm{var}(X) = \frac{2}{\mu^2}.$$

Substituting into the Pollaczek-Khinchin formula we have

$$\bar{W} = \frac{\lambda}{\mu^2(1-\rho)} = \frac{\rho}{\mu(1-\rho)}$$

5 marks for the distribution, 5 for the second moment and 5 marks for the result

d) Which service time distribution leads to the lowest delay for a given mean? What characteristic of this distribution ensures it has the lowest associated delay?

[15%]

Answer

The service time distribution that leads to the lowest delay for a given mean $\bar{X}$ is the deterministic service time, *i.e.* an $M/D/1$ queue. This is because it is the distribution with the lowest variance (the variance is zero) and therefore the second moment, $\langle X^2 \rangle = \bar{X}^2 + \mathrm{var}(X)$ is as low as it can be for this distribution (the variance cannot be negative) which in turn means the average delay is as low as it can be. There is no other distribution with a variance of zero.

5 marks for the right answer, and 10 marks for the fact that it is the lowest variance.

2. This question focuses on the Erlang loss system

a) Consider the Erlang B formula,

$$B(m,a) = \frac{\frac{a^m}{m!}}{\sum_{n=0}^{m} \frac{a^n}{n!}}.$$

What is the significance of the result of this formula? What do a and m stand for?

[20%]

Answer

The formula gives the blocking probability for the Erlang loss system. $a = \frac{\lambda}{\mu}$ is the 'offered load', a dimensionless constant measured in Erlangs that gives the system load and m is the number of servers and the capacity of the system.

5 for a and m, 10 for the significance

b) What problems might you have when trying to compute the result of this formula on either a calculator or a computer? How would you seek to solve these problems?

[25%]

Answer

For large values of m there will be overflow problems for computing a^m and $m!$ This problem can be solved by computing the result in log space and then exponentiating. To do this you use the log gamma function for computing the factorial term (if they mention the use of curves to solve the problems half marks for the solution part).

***15 for identifying the problem and 10 for identifiying the solution.*0**

c) The steady state distribution for the Erlang loss system is given by

$$P(k) = \frac{\frac{a^k}{k!}}{\sum_{n=0}^{m} \frac{a^n}{n!}}.$$

If the number of servers is very large ($m >> a$) what will the average number of customers in the system be?

Hint: consider what happens in the system as $m \to \infty$.

[25%]

Answer

As the question states we should consider the limit as $m \to \infty$. The steady state distribution for Erlang Loss is a truncated Poisson. In the limit $m \to \infty$ (which is an $M/M/\infty$ queue), the truncated Poisson becomes a Poisson distribution,

$$\sum_{n=0}^{\infty} \frac{a^n}{n!} = \exp(a),$$

so we have

$$P(k) = \frac{a^k}{k!}\exp(-a)$$

and the mean of this distribution is $\frac{1}{a}$.

15 for determining that the system's steady state approximates a Poisson for $m >> a$ (either by stating its originally a truncated Poisson, or by producing the result for the infinte sum becoming an exponential.) 10 for knowing (or proving!) that the mean of a Poisson is $\frac{1}{a}$.

d) The Erlang B formula is associated with the Erlang loss queueing system. Under what conditions does the steady state distribution exist for this queueing system? What is the form of this steady state distribution when a becomes much larger than m and what is the resulting effect on the carried load?

[30%]

Answer

The steady state *always exists*. In some systems steady state doesn't exist when the offered load, a, is higher than the number of servers, m, however in the Erlang loss system the steady state will converge towards $P(k) = 1$ for $k = m$ and $P(k) = 0$ otherwise. In this case most packets are rejected and the carried load becomes very low. The steady state exists because the system is lossy.

5 marks for noting it always exists, 10 for giving the form of the steady state and 5 for stating the carried load will be small.

0

3. Annotate and hand in the Erlang C curves with your answer to this question.

United Telecom has just completed the purchase of several local telecommunications companies, each with their own regional support teams. There are three regional support teams. The number of staff in each team and the call rate for each team's centre is given in the following table.

Region	X	Y	Z
m	12	15	8
customers/hour	50	70	25

In the questions that follow you should assume that the call centres can be accurately modelled as Erlang delay systems.

a) Given that the average holding time, τ, for support queries is 6 minutes, use the Erlang C curves provided to give, for each region, the probability of queueing, $C(m,a)$.

[30%]

Answer

The offered load for this region is $a = \lambda\tau$, we have a holding time of 6 minutes = 6/60 hours.

- Region X: $a_X = \frac{50\times 6}{60} = \frac{30}{6} = 5$ and $m = 12$, from Erlang C curves$C(12,5) = 0.006$
- Region Y: $a_Y = \frac{70\times 6}{60} = \frac{42}{6} = 7$ and $m = 15$, from Erlang C curves $C(15,7) = 0.006$
- Region Z: $a_Z = \frac{25\times 6}{60} = \frac{15}{6} = 2\frac{1}{2}$ and $m = 8$, from Erlang C curves $C(8,2.5) = 0.005$

Answers should also be marked on Erlang C curves.

10 for each probability of queueing.

b) The formula for the average queue length in the Erlang delay system is

$$\bar{N}_Q = \frac{\rho}{1-\rho}C(m,a)$$

where ρ is the queue utilization factor. Use Little's formula to derive $\bar{W}$, the average queueing time, and use it compute the expected queueing time for each call centre.

[20%]

Little's formula for the queue is $\bar{N}_Q = \lambda\bar{W}$. So this means that

$$\bar{W} = \frac{1}{\mu(1-\rho)}C(m,a) = \frac{1}{\mu\left(1-\frac{a}{m}\right)}C(m,a)$$

The queueing time for each centre is therefore

- Region X: $\bar{W} = \frac{1}{\frac{60}{6}\left(1-\frac{5}{12}\right)}C(12,5) = \frac{6}{35}\times 0.006 = 8.57\times 10^{-4}$hours or about 3.7 seconds.
- Region Y: $\bar{W} = \frac{1}{\frac{60}{6}\left(1-\frac{7}{15}\right)}C(15,7) = \frac{3}{16}\times 0.006 = 9.375\times 10^{-4}$ hours or 4.05 seconds.
- Region Z: $\bar{W} = \frac{1}{\frac{60}{6}\left(1-\frac{2.5}{8}\right)}C(8,2.5) = \frac{8}{55}\times 0.005 = 6.060\times 10^{-4}$ hours or 2.618 seconds.

5 for derivation of the formula and 5 for each answer.

c) Show that the average queueing time for customers that queue is given by

$$\bar{W}_Q = \frac{1}{\mu(1-\rho)}$$

and compute the average queue time for customers for each of the call centres described above.

[25%]

Answer

The total queueing time is given by

$$\bar{W} = C(m,a)\,\bar{W}_Q + (1 - C(m,a))\,\bar{W}_{\sim Q}$$

where

$$\begin{aligned} \bar{W}_Q &= \text{queueing time for customers that queue.} \\ \bar{W}_{\sim Q} &= \text{queueing time for customers that don't queue.} \end{aligned}$$

Since $\bar{W}_{\sim Q} = 0$ we can reorganise the formula as

$$\bar{W}_Q = \frac{\bar{W}}{C(m,a)}.$$

The queueing time for those that queue at each centre is therefore

- Region X: $\bar{W} = \frac{1}{\frac{60}{6}\left(1-\frac{5}{12}\right)} = \frac{6}{35}$ hour or 10.28 minutes.
- Region Y: $\bar{W} = \frac{1}{\frac{60}{6}\left(1-\frac{7}{15}\right)} = \frac{15}{80}$ hour or 11.25 minutes.
- Region Z: $\bar{W} = \frac{1}{\frac{60}{6}\left(1-\frac{2.5}{8}\right)}\frac{8}{55}$ hour or 8.73 minutes

10 for the derivation of the formula and 5 for each answer.

d) The company wishes to cut costs by merging the call centres to one location. Assuming that the total traffic level and the average holding time remain the same, how many staff will be needed to keep the probability of waiting at 0.006 or below?

[5%]

Answer

From the Erlang C curves, 26.

5 for the answer (should be indicated on the Erlang C curves as well).

e) Explain the difference in the number of staff required in the centralized call centre in comparison to the three original call centres. In your answer you should focus on the utilization factors in the two systems.

[20%]

Answer

The centralized system requires 9 fewer staff than the original three separate systems for approximately the same performance. The reason for this improved performance is the improved efficiency of the system. In the original system the average efficiency can be found from the utilization factors.

- Region X: $\rho_X = \frac{\lambda}{m}\tau = \frac{50\times 6}{12\times 60} = \frac{5}{12}$
- Region Y: $\rho_Y = \frac{\lambda}{m}\tau = \frac{70\times 6}{15\times 60} = \frac{7}{15}$
- Region Z: $\rho_Z = \frac{\lambda}{m}\tau = \frac{25\times 6}{8\times 60} = \frac{5}{16}$

so the average efficiency is $\frac{1}{3}\left(\frac{5}{12}+\frac{7}{15}+\frac{5}{16}\right) = 0.399$
For the centralised system $m = 26$, $\lambda = 50+70+25 = 145$ (by superposition of Poissons), and $\tau = 0.1$. So efficiency is

$$\rho = \frac{\lambda\tau}{m} = \frac{145\times 0.1}{26} = 0.557.$$

The second system is therfore more efficient which explains the reduction in numbers of staff required.

10 for the basic answer (i.e. the improved efficiency of the system) and 10 for the actual figures.

4. You are presented with the following portion of Tcl code designed for the network simulator NS2.

```
set node1 [$ns node]
set node2 [$ns node]
set node3 [$ns node]

$node1 set multiPath_ 1

set link1 [$ns duplex-link $node1 $node2 2Kb 0ms DropTail]
set link2 [$ns duplex-link $node2 $node3 2Kb 0ms DropTail]
set link3 [$ns duplex-link $node1 $node3 2Kb 0ms DropTail]

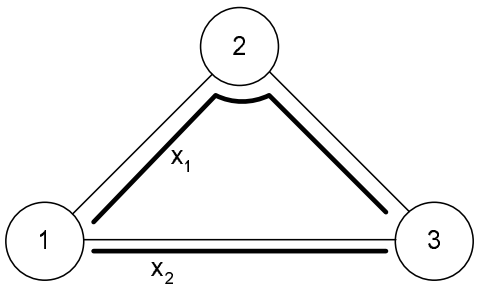
set src [new Agent/UDP]
$ns attach-agent $node1 $src
set sink [new Agent/Null]
$ns attach-agent $node3 $sink
```

a) Assuming that only the specified nodes exist, draw the network topology showing the associated routes.

[25%]

Answer

The `setlink` statements give the structure of the network and the `setsrc` and `setsink` statements along with the `setmultiPath` statment ensure that packets travel from `node1` to `node3` along all possible routes.



In the diagram above we have omitted the arrows that show this, they should either be shown or indicated in accompanying text.

10 marks for the correct structure and 15 marks for the correct routing information.

b) Packet arrivals occur as a Poisson distribution at `node1` with a rate of 2 per second. Packet lengths are distributed as an exponential distribution with a mean of 0.5 Kb. Assuming that packets are sent randomly along all possible routes with equal probability, use Kleinrock's independence assumption to estimate the steady state average number of packets at each link and the average delay for each route.

[25%]

Answer

Packets arrive as a Poisson with a rate $\lambda = 2$. They are sent randomly along the two routes with equal probaility. This means, via the decomposition property of Poisson processes, that the rate for each route is $\lambda = 2 \times 0.5 = 1$. The service time distribution is dicated by the length of the packet and the bandwidth of the route. The bandwidth (given in the code above) is 2Kb/s. Packet lengths are 0.5 Kb, so the service time distribution for all links will have $\mu = \frac{2}{0.4} = 4$ packets/s. Kleinrock's independence assumption tells us to treat each queue independently. The first route consists of two queues.

The delay is given by the formula for the $M/M/1$ queue which is

$$\bar{T} = \frac{1}{\mu - \lambda} = \frac{1}{4-1} = \frac{1}{3}s.$$

There are two queues in the first route so the total delay is $\frac{2}{3}s$.

The calculation is the same for the second route, but there there is only one queue (so in fact you don't need Kleinrock's assumption — see later question part) and the delay in the second route is $\frac{1}{3}s$. ***5 for decomposition of Poissons 10 for the correct formula and 5 for the correct answer for each route.***

c) One of the routes in the network contains two queues. Explain how the second queue's size, as perceived by an arriving packet, will be related to the arriving packet's length. How will this waiting time be affected for longer packets arriving at the second queue?

[20%]

Answer

The longer packets will arrive at the second queue to find a shorter queue than the shorter packets. Or in other words queue length at the second queue are anti correlated with packet lengths. This is because, while the longer packet was being served at the first queue there is time for the second queue to empty out. This is analogous to a slow moving truck driver who will generally see an open road in front of him.

10 marks for correctly saying that the queue will be shorter for longer packets (or vice versa). 10 Marks for saying the longer packet will wait less time at the second queue.

d) For each route in the network consider the implications of Kleinrock's independence assumption. For which route would you expect the assumption to lead to more accurate results? Describe the reasoning behind your answer.

[20%]

Answer

The second route which goes directly from node 1 to node 3 would be much more accurate. This is because the assumptions behind Kleinrock are valid for this route. The packet lengths are uncorrelated with the arrival times. For the first route (via node 2) the packet lengths become correlated through passing through the queue at node 2.

10 marks for the answer and 10 marks for the reasoning (that independence holds for the second route.)**0**

e) Typically, in a large network, analytical modelling of the system will not be possible. In practice we often turn to simulation for modelling of such systems. Why might the sample-based approximations to the system's characteristics we obtain from these simulations be misleading?

[10%]

Answer

If the simulation is not run long enough it may not reach it's steady state. It may even be hard to determine whether the system has a steady state. If samples from the system are taken too close together in time then they are not truly independent and sample based estimates will be incorrect. ***5 marks for steady state and 5 marks for independence. Will also award 5 marks to answers that state you need to obtain a lot of samples to get accurate results, but to get 10 you need to have both the first points as well.*****0**

END OF QUESTION PAPER

Registration Number _____

